

Md Abrar Jahin, CSCATM

Email(s): jahin@usc.edu; jahin@isi.edu

Phone: (+1) 213-245-5908

Work: ISI 718, 4676 Admiralty Way, Marina del Rey, CA 90292

[GitHub](#)

[LinkedIn](#)

[Personal Website](#)

[Google Scholar](#)

[ResearchGate](#)

[ORCID](#)

Education

University of Southern California (USC)

Ph.D. in Computer Science

- Fellowships: Capital One Fellow (2026–2027); Viterbi Graduate School Fellow (2025–2026)
- Co-advised by [Prof. Craig Knoblock](#) and [Prof. Jay Pujara](#)

Los Angeles, CA, USA

Aug 2025 – Present

Khulna University of Engineering & Technology (KUET)

B.Sc. Eng. in Industrial & Production Engineering

CGPA: 3.83/4.00 (Top 5% of class)

- Dean's award: 2018-2019, 2019-2020, 2020-2021
- Thesis title: Supply Chain Backorder Prediction Using Interpretable Hybrid Quantum–Classical Neural Network [[Thesis Presentation](#)] [Supervisor: [Dr. Md. Saiful Islam](#)]
- Developed the first-ever open-source LaTeX template for B.Sc. Undergrad Thesis of KUET [[Template](#)]
- First Highly Commended Research Awardee from the Global Undergraduate Awards (2025) in KUET's history
- [Google Knowledge Panel](#) of Md Abrar Jahin

Khulna, Bangladesh

Nov 2018 – Mar 2024

Research Interests

† Efficient Deep Learning (DL)

- *Geometric & Equivariant Deep Learning, Spiking Neural Networks (SNN), Kolmogorov-Arnold Networks (KAN), Physics-informed Neural Networks (PINN), Neural Operators*

† Graph Representation Learning & Neurosymbolic AI

- *Graph Neural Networks (GNN); Temporal / Dynamic Graphs*
- *Knowledge Graphs, Ontologies & Reasoning (Semantic Web)*
- *Scientific & Automated Knowledge Discovery*

† Quantum Machine Learning (QML)

- *Equivariant / Geometric QML, Neural Quantum States (NQS) for Many-Body Physics*

† Trustworthy AI

- *Explainable AI (XAI)*
- *Uncertainty Quantification*
 - ◇ *Conformal Prediction*
- *Certified / Verifiable ML*

† Foundation Models & Agentic AI

- *Multi-Agent & Human-AI Knowledge Systems*

† Reinforcement Learning (RL)

- *Inverse RL, Imitation Learning, Human-in-the-loop RL*

Research Experiences

❖ AI Researcher

Center of Knowledge Graphs, USC Information Sciences Institute (ISI), Marina del Rey, CA, USA

Supervisors: [Prof. Craig Knoblock](#) and [Prof. Jay Pujara](#)

Research projects:

[1] Modeling Sparse and Heterogeneous Geochemistry Data to Accelerate Critical Mineral Discovery (STTR Phase II) [Funded by **DARPA/USGS**; \$759,346; Jan 2026 – Jul 2027](Collaborators: [Inferlink](#), [Jataware](#))

- Developing a geochemistry knowledge base to integrate sparse mining literature data
- Automating article discovery, method extraction, and structured table understanding
- Supporting USGS critical mineral resource estimation

[2] Enhancing Scalability of Knowledge Graphs for Enterprise Defense Solutions (STTR Phase I; Apr 2026 - Jun 2026) [Funded by **U.S. Air Force Research Laboratory (AFRL)**, led by [SIMBA Chain, Inc.](#)]

[3] FinSemKG: A Semantic Mediation Layer for Grounded, Auditable LLM Question Answering over Financial Tables [Funded by the **USC–Capital One Center for Responsible AI Decision Making in Finance (CREDIF)**; Project Award: \$84,000; Aug 2026–Aug 2027; Role: Capital One Fellow (50% Research Assistant); PI: Prof. Craig Knoblock]

- Collaborators: [Dr. Anoop Kumar](#), [Dr. Wenqing Zheng](#), [Dr. Lev Telyatnikov](#)

[4] Temporal Knowledge Graphs for Multi-Hop Pattern Recognition

Aug 2025 – Present

❖ Machine Learning Researcher

Upwork, Freelance (Remote)

Jan 2025 – Jul 2025

■ Awarded the ‘Rising Talent’ badge on Upwork within six months of joining, in recognition of my in-demand skills, strong performance, and impressive professional background.

❖ Lead Researcher

Mar 2023 – Aug 2025

Advanced Machine Intelligence Research Lab (AMIRL), American International University-Bangladesh (AIUB)

Roles: Research Assistant (Mar 2023 - Dec 2023), Researcher (Dec 2023 - Feb 2024), Lead Researcher (May 2024 - Aug 2025)

Supervisors: [Prof. M. F. Mridha](#) and [Prof. Nilanjan Dey](#)

◆ Collaborators: [Prof. R. Simon Sherratt](#) (IEEE Fellow, IET Fellow), [Prof. Jungpil Shin](#), [Prof. Yuichi Okuyama](#), [Prof. Zeyar Aung](#), [Prof. Yutaka Watanobe](#), [Prof. Md. Rashedul Islam](#), [Prof. Md. Jakir Hossen](#), [Prof. Md. Abdul Hamid](#), [Prof. Muhammad Mostafa Monowar](#)

■ Published **10+ WoS Q1** journal articles and **CORE-ranked** conference articles (concentration: DL, QML, GNN, XAI, conformal prediction, human-in-the-loop, NLP, and operations research).

❖ Visiting Researcher (VR)

Mar 2024 – Mar 2025

Physics and Biology Unit, Okinawa Institute of Science and Technology Graduate University (OIST), Japan

Supervisor: [Prof. Jonathan Miller](#) [BS (Yale); PhD Biology (Cambridge); PhD Physics (Caltech)]

Research project: Evolutionary Dynamics of Strongly Conserved Sequences in Vertebrates and Insects [[Code Repository](#)]

■ [[FY2023 Annual Report](#)] [[OIST Affiliation](#)]

❖ Visiting Research Student (VRS)

Feb 2023 – Feb 2024

Physics and Biology Unit, OIST, Japan

Supervisor: [Prof. Jonathan Miller](#) [BS (Yale); PhD Biology (Cambridge); PhD Physics (Caltech)]

Research project: Evolutionary Dynamics of Strongly Conserved Sequences in Vertebrates and Insects [[Certificate](#)]

Collaborators: [Dr. Lucia Žifčáková](#), [Dr. Priscila Do Nascimento Biller](#), [Dr. Zdenek Lajbner](#), and [Dr. Reuven Pnini](#)

■ Critically analyzed and visually represented all potential combinations of inter-gap segments (IGS), ancestral repeats (ARs), and contiguous mismatched ARs in human/mouse and human/gorilla genome alignments, focusing on both DNAs and repetitive sequences.

■ Successfully replicated the findings of the neutral indel model proposed by [Lunter, Pointing, and Hein \(2006\)](#).

❖ Research Lead

May 2022 – Mar 2023

Research Camp 02, Scholarship School BD, Bangladesh

Supervisor: [Dr. Mohammad Arafat Hussain](#) (Post-doctoral Research Fellow at Image, Informatics & Intelligence Research Lab, Harvard Medical School; PhD in Biomedical Eng., UBC Canada; MSc in Biomedical Eng., UBC)

■ Led the research team of 17 fellow researchers as a co-first author on a project titled “[Ultrasound-Based AI for COVID-19 Detection: A Comprehensive Review of Public and Private Lung Ultrasound Datasets and Studies](#)”.

■ Contributed to writing the original manuscript, software implementation, and data curation, and served as a corresponding author for the entire communication with the journal.

❖ Research Intern (RI)

Oct 2021 – Mar 2022

Physics and Biology Unit, OIST, Japan

Supervisor: [Prof. Jonathan Miller](#) [BS (Yale); PhD Biology (Cambridge); PhD Physics (Caltech)]

■ Awarded a full-funded scholarship with a daily allowance of JPY 2400 per working day [[Offer Letter](#)] [[RI Agreement](#)] [[Internship Certificate](#)] (acceptance rate: **14%**)

■ Tracked erroneous out-of-bound PCS coordinates generated by [Nash and Lenhard \(2018\)](#), utilizing R and Bedtool, and resolved complexity issues using Python 3.10.

■ Demonstrated shell scripting and parallel computing proficiency on the HPC Deigo cluster.

■ Conducted research on perfectly conserved sequence (PCS) length distributions of UCSC 44 pairwise genome sequences.

■ Analyzed quantile kurtosis of PCS lengths proposed by [Nash and Lenhard \(2018\)](#) and identified a ‘knee’ in the PCS distributions of the heavy-tailed region.

■ Optimized Nash and Lenhard’s 3 R scripts for PCS generation, quantile kurtosis analysis, and genomic regulatory blocks (GRBs), reducing time and memory complexity. Successfully reproduced PCS coordinates following UCSC format and fixed genome coordinate-related errors in R’s Bioconductor package.

❖ Research Intern

May 2021

UiT - The Arctic University of Norway

Supervisors: [Prof. Aleksander Pedersen](#), [Prof. Rune Dalmo](#), [Ghada Bouzidi](#)

Internship Mentor: [Prof. Per Arne Sundsbø](#)

- Conducted comprehensive data and statistical analysis on the Narvik road dataset as part of the DIT4BEARs Smart Road project. [\[Internship Report\]](#) [\[Project Source Code\]](#) [\[Certificate\]](#)
- Designed, implemented, and evaluated ML models that successfully identified six road states, addressing the challenges of winter weather conditions in the Barren Euro-Arctic region.
- Proposed a novel safety metric and utilized Ridge, Lasso, Elastic Net, Linear Regression, and XGBRegressor to forecast its values.

Publications

I have **730+** citations according to Google Scholar as of September 14, 2026 (**h-index = 13**, **i10-index = 14**)

* Denotes co-first authorship.

Journal Articles

- [J-20] **Jahin, M. A.**, Naife, S. A., Saha, A. K., & Mridha, M. F. (2026). AI in Supply Chain Risk Assessment: A Systematic Literature Review and Bibliometric Analysis. *SN Business & Economics*. 
- [J-19] Rahman, M. M. *, **Jahin, M. A. ***, Islam, M. S., & Mridha, M. F. (2026). Optimizing Container Loading and Unloading through Dual-Cycling and Dockyard Rehandle Reduction Using a Hybrid Genetic Algorithm. *Journal of Marine Science and Technology*. 
- [J-18] **Jahin, M. A.**, Soudeep, S., Mridha, M. F., Monowar, M. M., & Hamid, M. A. (2026). Physics-Informed Graph Neural Networks for Transverse Momentum Estimation in CMS Trigger Systems. *Computer Physics Communications*, 325, 110167. 
- [J-17] Nasif, S. M. *, **Jahin, M. A. ***, & Mridha, M. F. (2026). Reinforcement-Guided Hyper-Heuristic Hyperparameter Optimization for Fair and Explainable Spiking Neural Network-Based Financial Fraud Detection. *Knowledge-Based Systems*, 341, 115814. 
- [J-16] **Jahin, M. A.**, Masud, M. A., Mridha, M. F., Dey, N., & Aung, Z. (2026). Quantum Rationale-Aware Graph Contrastive Learning for Jet Discrimination. *TMLR*. 
- [J-15] **Jahin, M. A.**, Mridha, M. F., Dey, N., & Hossen, M. J. (2026). Human-in-the-Loop Feature Selection Using Interpretable Kolmogorov-Arnold Network-based Double Deep Q-Network. *IEEE Open Journal of Computer Society*, 7, 389–403. 
- [J-14] Uddin, M. K., Islam, M. S., **Jahin, M. A.**, Irfan, M. T. H., Seam, M. S. I., & Mridha, M. F. (2026). Designing Cellular Manufacturing Systems in the Presence of Alternative Process Plans. *IET Collaborative Intelligent Manufacturing*, 8(1), e70079. 
- [J-13] Uddin, M. K., Islam, M. S., **Jahin, M. A.**, Seam, M. S. I., & Mridha, M. F. (2026). Solving Generalized Grouping Problems in Cellular Manufacturing Systems Using a Network Flow Model. *Journal of Intelligent Manufacturing and Special Equipment*, 7(2), 264–293. 
- [J-12] **Jahin, M. A.**, Naife, S. A., Lima, F. T. J., Mridha, M. F., & Hossen, M. J. (2025). Predicting Male Domestic Violence Using Explainable Ensemble Learning and Exploratory Data Analysis. *Discover Applied Sciences*, 8(26). 
- [J-11] **Jahin, M. A.**, Masud, M. A., Mridha, M. F., Aung, Z., & Dey, N. (2025). KACQ-DCNN: Uncertainty-Aware Interpretable Kolmogorov-Arnold Classical-Quantum Dual-Channel Neural Network for Heart Disease Detection. *Computers in Biology and Medicine*, 197(A), 110976. 
- [J-10] **Jahin, M. A. ***, Shahriar, A. *, & Amin, M. A. (2025). MCDNF: Supply Chain Demand Forecasting via an Explainable Multi-Channel Data Fusion Network Model. *Evolutionary Intelligence*, 18(3), 66. 
- [J-9] Morshed, A., Shihab, A. A., **Jahin, M. A. ***, Nahian, M. J. A., Sarker, M. M. H., Wadud, M. S. I. *, Uddin, M. I., Siraji, M. I., Anjum, N., Shristy, S. R., Rahman, T., Khatun, M., Dewan, M. R., Hossain, M., Sultana, R., Chakma, R., Emon, S. B., Islam, T., & **Hussain, M. A. *** (2025). Ultrasound-Based AI for COVID-19 Detection: A Comprehensive Review of Public and Private Lung Ultrasound Datasets and Studies. *Multimedia Tools and Applications*, 84, 41615–41669. 
- [J-8] **Jahin, M. A.**, Masud, M. A., Suva, M. W., Mridha, M. F., & Dey, N. (2025). Lorentz-Equivariant Quantum Graph Neural Network for High-Energy Physics. *IEEE Transactions on Artificial Intelligence*, 6(12), 3195–3206. 
- [J-7] **Jahin, M. A.**, Shovon, M. S. H., Mridha, M. F., Islam, M. R., & Watanobe, Y. (2024). A hybrid transformer and attention based recurrent neural network for robust and interpretable sentiment analysis of tweets. *Scientific Reports*, 14(1), 24882. 
- [J-6] Shahriar, H. *, Islam, M. S., **Jahin, M. A. ***, Ridoy, I. A., Prottoy, R. R., Abid, A., & Mridha, M. F. (2024). Exploring Internet of Things Adoption Challenges in Manufacturing Firms: A Delphi Fuzzy Analytical Hierarchy Process Approach. *PLoS ONE*, 19(11), e0311643. 
- [J-5] Saha, A. K. *, **Jahin, M. A. ***, **Rafiquzzaman, M.**, & Mridha, M. F. (2024). Ergonomic Design of Computer Laboratory Furniture: Mismatch Analysis Utilizing Anthropometric Data of University Students. *Heliyon*, 10(14). 
- [J-4] Ahmad, K. *, Islam, M. S., **Jahin, M. A. ***, & Mridha, M. F. (2024). Analysis of Internet of things implementation barriers in the cold supply chain: An integrated ISM-MICMAC and DEMATEL approach. *PLoS ONE*, 19(7), e0304118. 

- [J-3] **Jahin, M. A.**, Shovon, M. S. H., Shin, J., Ridoy, I. A., & Mridha, M. F. (2024). Big Data - Supply Chain Management Framework for Forecasting: Data Preprocessing and Machine Learning Techniques. *Archives of Computational Methods in Engineering*, 31(6), 3619–3645. 
- [J-2] **Jahin, M. A.**, & **Talapatra, S.** (2024). A Natural Language Processing-Based Classification and Mode-Based Ranking of Musculoskeletal Disorder Risk Factors. *Decision Analytics Journal*, 11, 100464. 
- [J-1] **Jahin, M. A.**, Shovon, M. S. H., Islam, M. S., Shin, J., Mridha, M. F., & Okuyama, Y. (2023). QAmplifyNet: Pushing the boundaries of supply chain backorder prediction using interpretable hybrid quantum-classical neural network. *Scientific Reports*, 13(1), 18246. 

Under-Review Journal/Conference Articles

- [U-18] **Jahin, M. A.**, Fuad, T. R., Pujara, J., & Knoblock, C. A. (2026). Spectral Initialization and Scheduled Graph Smoothness for Uncertain Knowledge Graph Completion. Under review at **AAAI 2027**. 
- [U-17] Fuad, T. R., **Jahin, M. A.**, & Hussain, A. (2026). RINSE: Robust Target-Time Normality Estimation for Zero-Shot Graph Anomaly Detection. Under review at **AAAI 2027**. 
- [U-16] **Jahin, M. A.**, Akramuddoula, M., Parvez, M. R., Pujara, J., & Knoblock, C. A. (2026). Aethelred: Certifying Robustness of Invariance-Trained Graph Neural Network Explanations. Under review at **AAAI 2027**. 
- [U-15] **Jahin, M. A.**, Akramuddoula, M., Parvez, M. R., Pujara, J., & Knoblock, C. A. (2026). Certified Obfuscation for Privacy-Preserving Learning: Reconstruction Risk and a Keyed Rotation–Noise–Permutation Mechanism. Under review at **AAAI 2027**. 
- [U-14] **Jahin, M. A.**, Akramuddoula, M., Sanglap, N. T., Parvez, M. R., Pujara, J., & Knoblock, C. A. (2026). Refusal Is Layer-Distributed: A Causal Score for Where Safety Lives and an Adaptive Ablation That Uses It. Under review at **AAAI 2027**. 
- [U-13] Hasan, S. M. M., **Jahin, M. A.**, Parvez, M. R., Hossen, M. J., & Mridha, M. F. (2026). Conformance-Aware Repair of HTN Planning Domains: The Precision Gap and How to Close It. Under review at **AAAI 2027**. 
- [U-12] **Jahin, M. A.**, Kirttania, T., Ghosh, L., & Parvez, M. R. (2026). PhyloAgent: Contrastive Failure Localization via Successful-Trace Alignment in Multi-Agent LLM Systems. Under review at **AAAI 2027**. 
- [U-11] **Jahin, M. A.**, Knoblock, C. A., & Pujara, J. (2026). FLINT: Feature Ranking with Arborescence Decoding for Semantic Table Interpretation. Under review at **AAAI 2027**. 
- [U-10] Hasan, S. M. M., **Jahin, M. A.**, Parvez, M. R., Hossen, M. J., & Mridha, M. F. (2026). Instance-BEq: Formal Faithfulness Certification for Structure-to-Instance Theorem Autoformalization. Under review at **AAAI 2027**. 
- [U-9] **Jahin, M. A.**, Fuad, T. R., Pujara, J., & Knoblock, C. A. (2026). Differentiable Koopman Operator for Contrastive Learning on Dynamic Graphs. Under review at **NeurIPS 2026**. 
- [U-8] **Jahin, M. A.**, Ahamed, S. S., Mridha, M. F., & Aung, Z. (2026). RELAY-Bench: A Benchmark and Portfolio Baseline for Incomplete MaxSAT Evaluation. Under review at the **NeurIPS 2026 Evaluations and Datasets Track**. 
- [U-7] Trishna, N. J., Mridha, M. F., Mahmud, S. M. H., **Jahin, M. A.**, & Aung, Z. (2026). High-Responsiveness Temporal Graph Neural Network for Early Prediction of Patient Deterioration in Intensive Care Units. Under review at **npj Cardiovascular Health**.
- [U-6] Trishna, N. J., **Jahin, M. A.**, Mridha, M. F., & Hossen, M. J. (2026). Deep Spatial-Relational Learning for Brain Tumor Localization Using CNN–GNN Architectures. Under review at **iScience**.
- [U-5] **Jahin, M. A.**, Abid, A., & Mridha, M. F. (2026). Quantum-Informed Contrastive Learning with Dynamic Mixup Augmentation for Class-Imbalanced Expert Systems. Under review at **Neurocomputing**. 
- [U-4] **Jahin, M. A.***, Soudeep, S.*, Mridha, M. F., & Dey, N. (2026). Interpretable Soybean Leaf Disease Classification with a Lightweight MobileNetV2-GraphSAGE Network. Under review at **Signal, Image and Video Processing**. 
- [U-3] Islam, M. A., **Jahin, M. A.**, Mridha, M. F., & Dey, N. (2026). A Unified Framework for Evaluating and Enhancing the Transparency of Explainable AI Methods via Perturbation-Gradient Consensus Attribution. Under review at **Neural Networks**. 
- [U-2] Soudeep, S.*, **Jahin, M. A.***, & Mridha, M. F. (2026). Interpretable Dynamic Graph Neural Networks for Small Object Detection and Tracking in Traffic Surveillance. Under review at **Image and Vision Computing**. 
- [U-1] **Jahin, M. A.**, Mridha, M. F., Aung, Z., Dey, N., & **Sherratt, R. S.** (2026). TriQXNet: Forecasting Dst Index from Solar Wind Data Using an Interpretable Parallel Classical–Quantum Framework with Uncertainty Quantification. Under review at **Quantum Machine Intelligence**. 

Conference Articles and Book Chapters

- [C-11] **Jahin, M. A.**, Knoblock, C. A., & Pujara, J. (2026, October 25–26). ArticleMiner: Ontology-Guided Knowledge Graph Construction from Scientific Publications. *6th International Workshop on Scientific Knowledge: Representation, Discovery, and Assessment (Sci-K @ ISWC'26)*, co-located with the 25th International Semantic Web Conference (ISWC). Bari, Italy.  [**CORE A**, Acceptance Rate: 14/28=50%]

- [C-10] **Jahin, M. A.**, Fuad, T. R., Pujara, J., & Knoblock, C. (2026, November 20–22). ChronoSpike: An Adaptive Spiking Graph Neural Network for Dynamic Graphs. *Proceedings of the Fifth Learning on Graphs Conference (LOG 2026)*. Northeastern University, Boston, MA, USA.
- [C-9] **Jahin, M. A.**, & Parvez, M. R. (2026, October 24–29). GUI-Primitives: Diagnosing Spatial Reasoning Failures in Vision-Language GUI Grounding. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP'26 Main)*. Budapest, Hungary. [**CORE A***, Acceptance Rate: 2719/17669=15.4%]
- [C-8] Masud, M. A., **Jahin, M. A.**, & Hasan, M. (2026, November 12–15). HeteRo-Select: Informativeness as the Participation Driver in Heterogeneous Federated Learning. *2026 IEEE International Conference on Data Mining (ICDM'26)*. Shenyang, China. [**CORE A***, Acceptance Rate: 392/1966=19.94%]
- [C-7] **Jahin, M. A.**, Knoblock, C. A., & Pujara, J. (2026, August 18–20). A Weisfeiler–Leman Characterization of Global-Attention Graph Transformers for Mixed-Integer Linear Programs. In *Proceedings of the 2nd Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS'26), Proceedings of Machine Learning Research (PMLR)*, 334(2), 123–142. Northeastern University, Boston, MA, USA. [**Lightning Oral**] [**Poster**]
- [C-6] **Jahin, M. A.**, Soudeep, S., Mridha, M. F., Fahad, N., & Hossen, M. J. (2025, October 9–13). DyCAF-Net: Dynamic Class-Aware Fusion Network. In *Proceedings of the 12th IEEE International Conference on Data Science and Advanced Analytics (DSAA'25)*. Birmingham, UK. [**CORE A**, Acceptance Rate: 85/226=37.61%]
- [C-5] **Jahin, M. A.**, Fuad, T. R., Mridha, M. F., Fahad, N., & Hossen, M. J. (2025, October 19). AdeptHEQ-FL: Adaptive Homomorphic Encryption for Federated Learning of Hybrid Classical-Quantum Models with Dynamic Layer Sparing [Poster Presentation]. 1st International Workshop on Biomedical Image and Signal Computing for Unbiasedness, Interpretability, and Trustworthiness (**BISCUIT @ ICCV'25**). In *2025 IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, Honolulu, Hawaii. [**CORE A***] [**Poster**]
- [C-4] **Jahin, M. A.**, Soudeep, S., Aditta, A. R., Mridha, M. F., Fahad, N., Hossen, M. J. (2025, August 16). In Ma, Y., et al. (Eds), *Vision Transformers for End-to-End Quark-Gluon Jet Classification from Calorimeter Images*. Generalizing from Limited Resources in the Open World (**GLOW @ IJCAI'25**), Montreal, Canada. *Communications in Computer and Information Science: Vol. 2640, pp. 135–150*. Springer. [**CORE A***] [**Poster**]
- [C-3] **Jahin, M. A.***, Soudeep, S.*, Farid, F. A., Mridha, M. F., Kabir, R., Islam, M. R., & Karim, H. A. (2025, July 1–4). CAGN-GAT Fusion: A Hybrid Contrastive Attentive Graph Neural Network for Network Intrusion Detection. In H. Fujita, Y. Watanobe, M. Ali, & Y. Wang (Eds), *Advances and Trends in Artificial Intelligence. Theory and Applications. 38th International Conference on IEA/AIE*, Kitakyushu, Japan. (Lecture Notes in Computer Science: Vol. 15707, pp. 415–428). Springer. [**CORE C**, Acceptance Rate: 89/130=68.46%]
- [C-2] Žifčáková, L., **Jahin, M. A.**, & Miller, J. (2022, December 13–15). *Perfectly conserved sequences (PCS) between human and mouse are significantly enriched for exonic small proteins*. Bioinformatics and Computational Biology Conference (**BBCC'22**), Virtual. [**Poster**]
- [C-1] Žifčáková, L., & **Jahin, M. A.** (2023, July 23–27). *Perfectly conserved sequences (PCS) between human and mouse are significantly enriched for small-protein coding sequence*. Society for Molecular Biology and Evolution (**SMBE'23**), Ferrara, Emilia-Romagna, Italy. [**Poster**]

Research Internship Report

- [R-1] **Jahin, M. A.**, & Krutsylo, A. (2021). DIT4BEARs Smart Roads Internship (arXiv:2107.06755). arXiv.

Grant/Funding

Competitive Research Funding | AI

- | | |
|--|----------------|
| [1] University of Aizu (Japan) (×2) for [J-1] & [J-3] <i>Research Sponsor</i> : Prof. Jungpil Shin | 2023 – 2024 |
| [2] Khalifa University (UAE) (×3) for [J-11], [U-3] & [U-10] <i>Research Sponsor</i> : Prof. Zeyar Aung | 2024 – Present |
| [3] Hamad Bin Khalifa University (Qatar) (×1) for [J-10] <i>Research Sponsor</i> : Prof. Md Al Amin | 2024 – 2025 |
| [4] University of Aizu (Japan) (×1) for [J-7] <i>Research Sponsor</i> : Prof. Yutaka Watanobe | 2024 |
| [5] Multimedia University (Malaysia) (×4) for [J-12], [J-13], [C-3], [C-4], [C-5], & [C-6] <i>Research Sponsors</i> : Prof. Md. Jakir Hossen & Prof. Hezerul Abdul Karim | 2025 |

Honors and Scholarships

- **Capital One Fellow 2026–27 | USC–Capital One CREDIF** 2026–2027
Selected as the designated fellow for FinSemKG, a grounded and auditable LLM question-answering project over financial tables; supported by a 12-month 50% research assistantship (project award: ≈ \$84,000) [**Bio**]
- **Highly Commended Research Awardee (x2 Articles) | The Global Undergraduate Awards (GUA)** 2025
Recognized in the top 10% globally in the Computer Science category for 2 undergraduate papers [U-14] & [C-6] and the only recipient worldwide in 2025 to receive dual commendations [**Honor**] [**Certificate**]
- **Viterbi Graduate School Fellowship | University of Southern California** 2025–2026
Prestigious first-year PhD fellowship awarded for exceptional academic promise. Includes \$44,290 annual stipend, full tuition (up to 36 units/year), transportation, student programming, health & dental insurance, and summer support.
- **Champion | CS50x Puzzle Day | Harvard University | Meta** 2025
Led a 4-person international team (Bangladesh, USA, India, and Pakistan) and solved advanced 9/9 puzzles (including Metapuzzle) [**Certificate**]
- **Student Researcher of the Year Award 2024 | KUET Research Society** 2024

	Published the highest number of high-impact research articles (Oct 2023 – Nov 2024) in KUET [Award]	
	□ MIT Solve 2024 Global Health Equity Challenge	2024
	Founded SpecX, an XAI-powered web app, for sentiment-driven disease profiling & specialist allocation [Solution]	
	□ Champion CS50x Puzzle Day Harvard University Meta	2024
	Led a 5-person international team (Bangladesh, USA, Morocco, and Pakistan) and solved advanced 9/9 puzzles (including Metapuzzle) [Certificate]	
	□ Dean's Award (x3) KUET	2023
	Received 3 Dean's awards in recognition of achieving annual GPAs ≥ 3.75 out of 4.00 in three consecutive years of undergraduate classes [Certificate]	
	□ NASA Space Apps Challenge Global Nominee	2023
	Led a 5-person team and forecast geomagnetic storms using hybrid deep neural networks from satellite data [Project] [30 seconds of glory video]	
	□ Finalist HONDA Y-E-S (Young Engineer and Scientist's) Award 2022	2023
	Awarded for being among the top 15 Bangladeshi young engineering undergrad student scientists [Award & certificate]	
	□ Junior Research Fellowship Bangladesh Space Research and Remote Sensing Organization	2022
	Nominated as the junior-most research fellow by SPARRSO among the other excellent 55 undergraduate researchers for the project titled "Disaster Damage Mitigation by Multispectral Remote Sensing Satellite Image Data Analysis: A Deep Learning Approach" [Project nomination] [Presentation video]	
	□ Qiskit Gold Level Translator English to Bengali	2021
	Translated 22101 and proofread 25375 words of IBM Qiskit's first-ever textbook, collaborating with the West Bengal and Bangladeshi Qiskit translator team of 36 members [Certificate]	
	□ Top 6 among 385 teams Entrepren Season-2: Crafting Visions	2021
	Developed a feasible and sustainable business canvas model for our Git and Jenkins integrated freelancing startup [Case solution] [Finalist]	
	□ Top 500 Google Android App Developer Challenge	2021
	Engineered a countdown timer app on <i>Android Studio</i> with <i>Jetpack Compose Beta</i> using Kotlin language [Source Code] [Google swags]	
	□ Global Champion Smart Roads Hackathon	2021
	Executed a 2-person team and devised an ML model to forecast winter road friction and was offered a 1-month research internship at UiT - <i>The Arctic University of Norway</i> [Project Page]	
	□ Winner ISCEA Ptak Prize Global SCM Case Competition	2020
	Led a 4-person team and achieved 70% scholarship for completing the course for the professional certification titled <i>Certified Supply Chain Analyst (CSCA)</i> [Case solution] [Certificate]	
	□ Champion CS50x Puzzle Day (Fall) Harvard University	2020
	Spearheaded a 4-person international team (Bangladesh, UK, Pakistan, and Mexico) and solved advanced 8/8 puzzles [Certificate]	
	□ Gold Honor Ranked top 3% IAAC	2020
	<i>International Astronomy & Astrophysics Competition</i> [Solution][Final round certificate]	
	□ Champion CS50x Puzzle Day (Spring) Harvard University	2020
	Led a 3-person international team by fostering diversity & inclusion (Bangladesh, Brazil, and India) and solved advanced 8/8 puzzles [Certificate]	
	□ International Asteroid Search Collaboration NASA	2020
	Administered a 4-person team and discovered 2 main belt asteroids by analyzing Pan-STARRS images using <i>Astrometrica</i> software [Certificate]	
	□ Gold Honor Ranked top 5% IYMC	2019
	<i>International Youth Math Challenge</i> [Solution][Final round certificate]	
	□ Government Board Merit-based Scholarship (x4)	2010 – 2018
	PSC (2010; 17th in Rajshahi Board; awarded for 2 years), JSC (2013; awarded for 2 years), SSC (2016; awarded for 2 years), HSC (2018; awarded throughout 4-year B.Sc.) Govt. Board Exams	
Teaching Experience	Intro to Programming with Python <i>Mini-Course Teacher, OIST, Japan</i> ♦ Topics covered: Intro, Anaconda, variables, lists, strings, control structures [Course materials & details] ♦ Fellow Teachers: Dr. Nicholas Wardhana and Dr. Jeremie Gillet	Jan 2022
Tutorials	Operations Research - Developed and presented the first-ever comprehensive Bengali online tutorials on Operations Research topics, facilitating 3rd-year IPE students. - Topics covered: Simplex Method, Two-Phase, Big M, Graphical Sensitivity Analysis, TORA. [YouTube Playlist] - Reference book: "Operations Research – An Introduction" by Professor Hamdy A. Taha.	Jan 2023
Leadership & Advisory	KUET Research Society <i>Co-founder & President</i>	Oct 2023 – May 2024

Roles	• Served as an Executive Committee Member and President of the Industrial Engineering and Management Unit • Supervising (as an alumnus) 6 groups of research students concentrating on ML-DL and computational fuzzy logic, fostering cross-departmental research collaboration • Teaching (as an alumnus) scientific research methodology, research ethics, and journal article formatting, meeting publication criteria, and acquiring funding for publications	
	USC CS Undergraduate Mentorship Program (Fall 2025) <i>Graduate Mentor</i> • Mentoring 2 undergraduate CS students at USC on research preparation, course planning, and pathways to graduate study through recurring one-on-one advising • Advising on faculty outreach, independent skill development, and research-focused internships and REU opportunities	<i>Oct 2025 – Present</i>
	IEEE Computer Society – Coastal Los Angeles Section <i>Treasurer @ Chapter C16 (CH06240)</i> • Managed chapter financial operations, including budgeting, expense review, record maintenance, and reporting to the Section Chair Prof. Mary Mehrnoosh Eshaghian-Wilner • Prepared and submitted annual financial statements (L-50) and ensured compliance with IEEE MGA policies and audit requirements	<i>Jan 2026 – Present</i>
Professional Service	Program Committee Member ☐ Sci-K co-located with ISWC 2026 ☐ AAAI 2027 ☐ ICDM 2026	<i>Jun 2026 – Present</i>
	Associate Editor ☐ Frontiers in Sustainable Food Systems (<i>Agricultural and Food Economics</i>)	<i>Jul 2026 – Present</i>
	Peer Reviewer Verified by Web of Science [ResearcherID] ☐ AAAI 2027 ☐ ICDM 2026 ☐ TMLR ☐ NeurIPS 2026 ☐ ICML 2026 (Gold Reviewer Award) ☐ PLOS One ☐ Quantum Machine Intelligence (Springer Nature) ☐ Information Systems (Elsevier) ☐ IEEE Transactions on Biomedical Engineering ☐ ICML 2025 Workshops: LXAI, AIW, DataWorld ☐ IEEE Transactions on Systems, Man and Cybernetics: Systems ☐ Scientific Reports (Nature Portfolio) ☐ ACM Transactions on Intelligent Systems and Technology ☐ Expert Systems with Applications (Elsevier) ☐ IEEE Access ☐ Neural Networks (Elsevier) ☐ Computers & Industrial Engineering (Elsevier) ☐ Multimedia Tools and Applications (Springer Nature) ☐ Engineering Applications of Artificial Intelligence (Elsevier) ☐ The Journal of Supercomputing (Springer Nature) ☐ Cluster Computing (Springer Nature) ☐ Journal of Contemporary African Studies (Taylor & Francis) ☐ Journal of Multidisciplinary Healthcare (Taylor & Francis) ☐ 7th European Conference on Industrial Engineering and Operations Management (Augsburg, Germany, July 2024)	<i>Jul 2023 – Present</i>
Professional Memberships	Competition Judge ☐ Venture Track, Innoventure Season 1, BUP Entrepreneurship and Innovation Club (May 2026) ☐ Entreprenet Season 3, North South University Young Entrepreneurs Society (December 2025)	
	† AAAI Student Member (Member # 649479) [mjahin@acm.org] † ACM Student Member (Member # 3553960, USC ACM Chapter) [mjahin@acm.org] [Website] † IEEE Graduate Student Member [Member # 96981120, Coastal Los Angeles Section] † IEEE Computational Intelligence Society Member † IEEE Computer Society	<i>Dec 2025 – Nov 2026</i> <i>Aug 2025 – Jul 2026</i> <i>Jan 2026 – Present</i> <i>Jan 2026 – Present</i> <i>Jan 2026 – Present</i>

† IEEE Communications Society Member
 † IEEE Young Professionals
 † Quantum Technical Community, IEEE Computer Society

Jan 2026 – Present
Jan 2026 – Present
Jan 2022 – Dec 2025

Skills

Programming

Advanced and Proficient in: Python, C/C++, R, SQL, SAS, Data Structure and Algorithm, Object Oriented Programming
Familiar with: Kotlin

Machine Learning: Classical Deep Learning, Quantum Machine Learning (Qiskit, PennyLane, TorchQuantum), XAI, NLP, DASK: Parallel Computing, Tensorflow, Keras, PyTorch, IBM Watson

Data Analysis and Optimization: Microsoft Excel, IBM SPSS, Minitab, TORA, Gurobi, Beautiful Soup, Biopython, Bioconductor, NetworkX, OpenCV

High-Performance Scientific Computing: Deigo & Saion Clusters (OIST), CKG & Eclair Clusters (USC ISI)

Operating System: Linux, Unix, Windows

Version Control: Git Bash, Github, GitLab

Product Design: AutoCAD 2021, SolidWorks 2022, Unity 2D

Referencing Software: Zotero, Mendeley

Reviewing & Bibliometric Analysis: Publish or Perish 8.0, Gephi, VOSviewer

Writing tools: L^AT_EX, Microsoft Word

Languages: American English, Bengali (Native), Hindi

Certifications

MITx: [CTL.SC4x: Supply Chain Technology and Systems](#) (Grade: 75%)

MITx: [6.431x: Probability - The Science of Uncertainty and Data](#) (Grade: 91%)

MITx: [CTL.SC0x: Supply Chain Analytics](#) (Grade: 83%)

MITx: [2.961.2x: Management in Engineering: Strategy and Leadership](#) (Grade: 77%)

HarvardX: [PH125.1x: Data Science: R Basics](#) (Grade: 83%)

Delftx: [UnixTx: Unix Tools: Data, Software and Production Engineering](#)

TAUx, IsraelX: [Unlocking Information Security: Part 1](#)

ISCEA: [Certified Supply Chain Analyst](#) (Grade: 88%)

Google: [IT Technical Support Fundamentals](#)

Google: [Crash Course on Python](#)

Google: [The Bits and Bytes of Computer Networking](#)

UCSanDiegoX: [DSE200x: Python for Data Science](#) (Grade: 89%)

Georgia Tech: [Speak English Professionally: In Person, Online & On the Phone](#)

IBM: [AI Chatbots without Programming](#)

IBM: [PY0101EN: Python 101 for Data Science](#)

Microsoft: [Introduction to Artificial Intelligence \(AI\)](#)